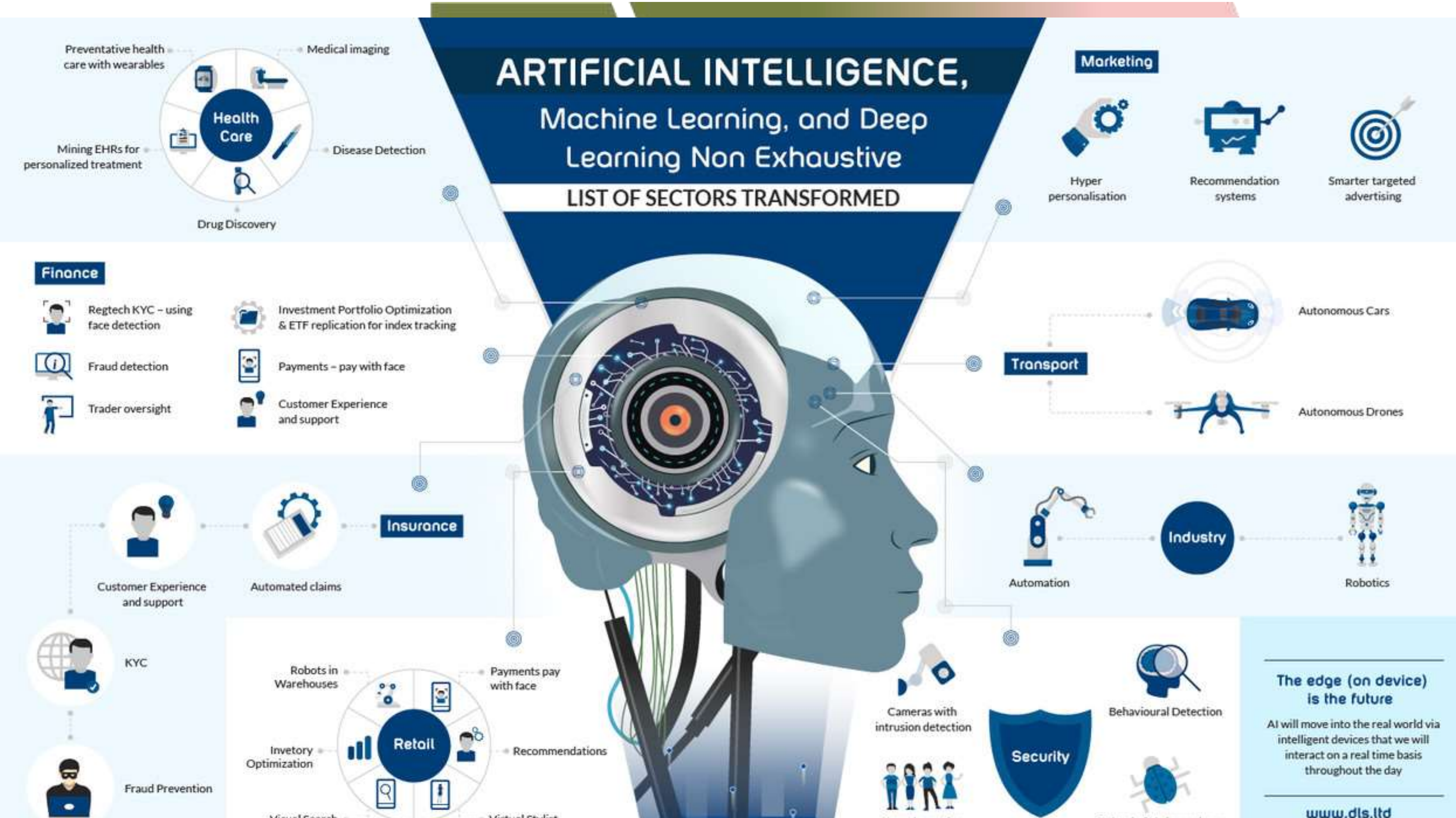
Deep Learning Crash Course

By : Vishwas Narayan


ARTIFICIAL INTELLIGENCE,
Machine Learning, and Deep Learning Non Exhaustive
LIST OF SECTORS TRANSFORMED
Health Care
Preventative health care with wearables
Medical imaging
Mining EHRs for personalized treatment
Disease Detection
Drug Discovery
Marketing
Hyper personalisation
Recommendation systems
Smarter targeted advertising
Finance
Regtech KYC – using face detection
Investment Portfolio Optimization & ETF replication for index tracking
Fraud detection
Payments – pay with face
Trader oversight
Customer Experience and support
Transport
Autonomous Cars
Autonomous Drones
Insurance
Customer Experience and support
Automated claims
KYC
Fraud Prevention
Industry
Automation
Robotics
Retail
Robots in Warehouses
Payments pay with face
Invetory Optimization
Recommendations
Security
Cameras with intrusion detection
Behavioural Detection
The edge (on device) is the future
AI will move into the real world via intelligent devices that we will interact on a real time basis throughout the day
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37


Natural language processing
Part 1: Tokenization
Coding TensorFlow

What you should Learn is to

- Build a groundbreaking intelligence just like humans through Deep Learning
- Build Neural Network with your approach and also make them make some sustainable Decision
- Understand and give your own approach to give a effective training for the deep learning model

What is the Difference between the Ai and others

Like

1. Deep Learning with ML and AI
2. Machine Learning with AI

Here never forget somehow you train to get the model

What you will learn?

- Loss Function and Optimizers
- Gradient Desce
- Neural Network

What

Microsoft Wor

A. M. Turing (1950) Computing Machinery and Intelligence. *Mind 49:* 433-460.

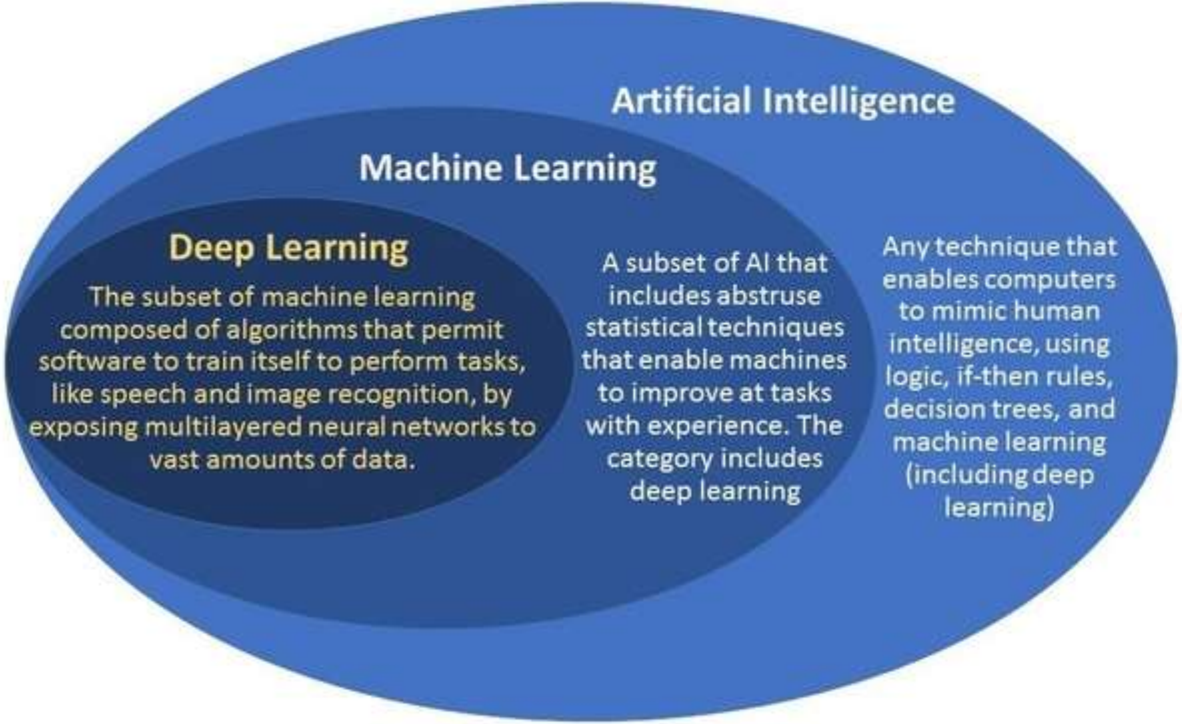
COMPUTING MACHINERY AND INTELLIGENCE

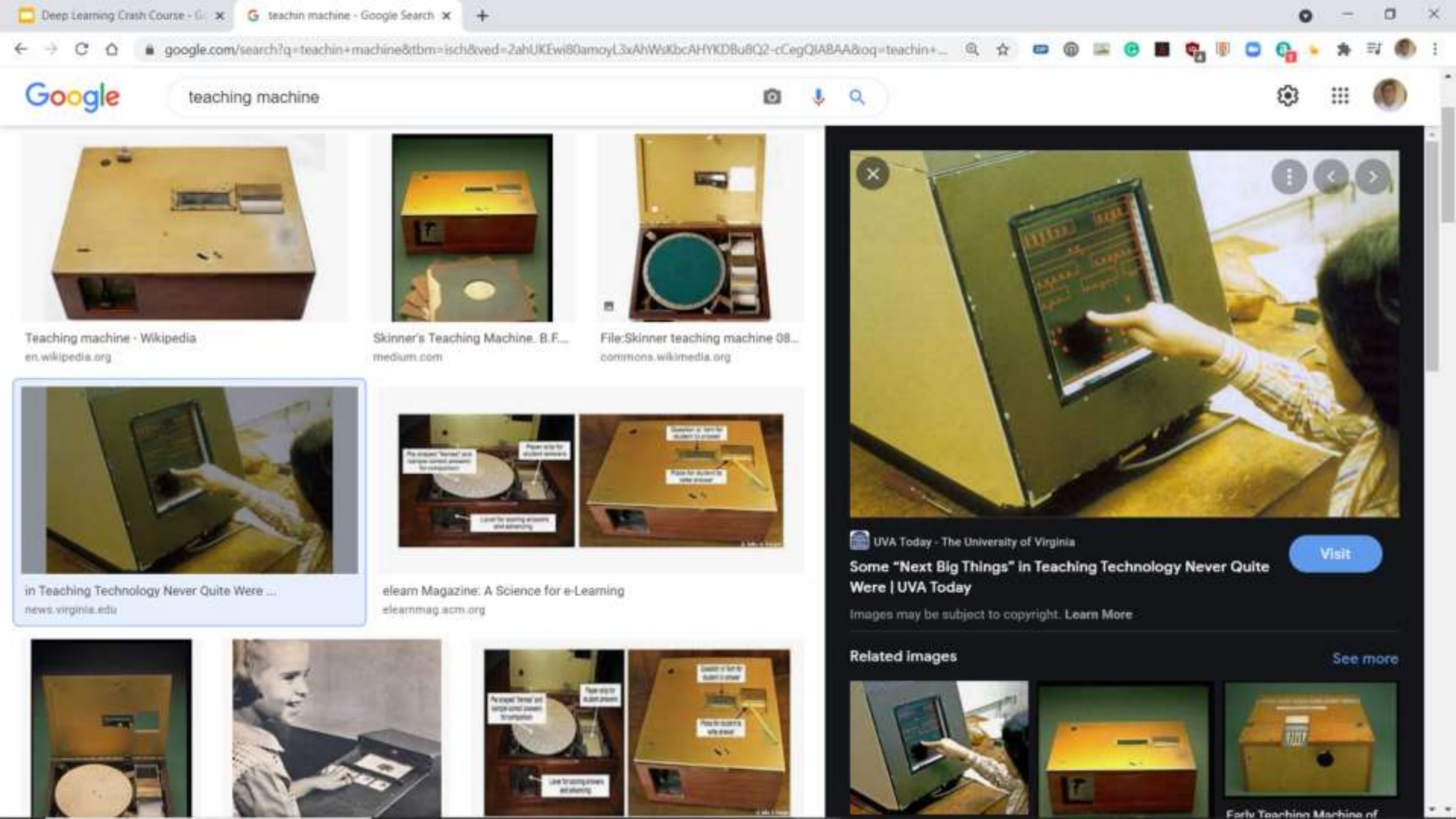
By A. M. Turing

1. The Imitation Game

I propose to consider the question, "Can machines think?" This should begin with definitions of the meaning of the terms "machine" and "think." The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous, If the meaning of the words "machine" and "think" are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, "Can machines think?" is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the 'imitation game." It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart front the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either "X is A and Y is B" or "X is B and Y is A." The interrogator is allowed to put questions to A and B thus:


Artificial Intelligence
Machine Learning
Deep Learning
The subset of machine learning composed of algorithms that permit software to train itself to perform tasks, like speech and image recognition, by exposing multilayered neural networks to vast amounts of data.
A subset of AI that includes abstruse statistical techniques that enable machines to improve at tasks with experience. The category includes deep learning
Any technique that enables computers to mimic human intelligence, using logic, if-then rules, decision trees, and machine learning (including deep learning)


Deep Learning Crash Course
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ALPHAGO
2017
BFI LONDON FILM FESTIVAL
WINNER
MILL VALLEY FILM FESTIVAL
Official Selection


THINK
$24,000
$77,147
$21,600
Who is Bram Stoker?
$ 17,973
$1,000

World's Top Companies Working on Self-Driving Cars

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THE BUSINESS OF ARTIFICIAL INTELLIGENCE

Exploring the mechanism of crashes with automated vehicles using statistical modeling approaches

Song Wang, Conceptualization, Investigation, Writing – original draft[#] and Zhixia Li, Conceptualization, Funding acquisition, Methodology, Supervision[#*]

Yanyong Guo, Editor

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Associated Data

▸ Data Availability Statement

Abstract

Go to:

Autonomous Vehicles (AV) technology is emerging. Field tests on public roads have been on going in several states in the US as well as in Europe and Asia. During the US public road tests, crashes with AV involved happened, which becomes a concern to the public. Most previous studies on AV safety relied heavily on assessing drivers' performance and behaviors in a simulation environment and developing automated driving system performance in a closed field environment. However, contributing factors and the mechanism of AV-related crashes have not been comprehensively and quantitatively investigated due to

What is Deep Learning

A Machine learning Technique that learn features and tasks directly from data.

Inputs are run through “Neural Networks”

Neural Network have hidden layers

Why Deep Learning?

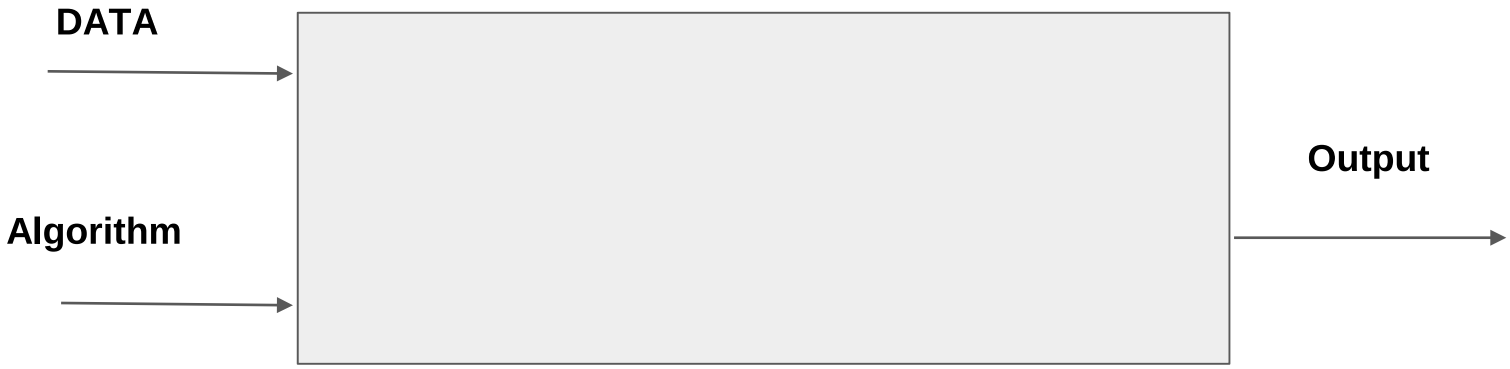
- Machines never get Fatigue.
- They need to get trained from the Human intelligence -
- They just fetch patterns.

In Deep Learning

Features can be le

What they Really mean to us?

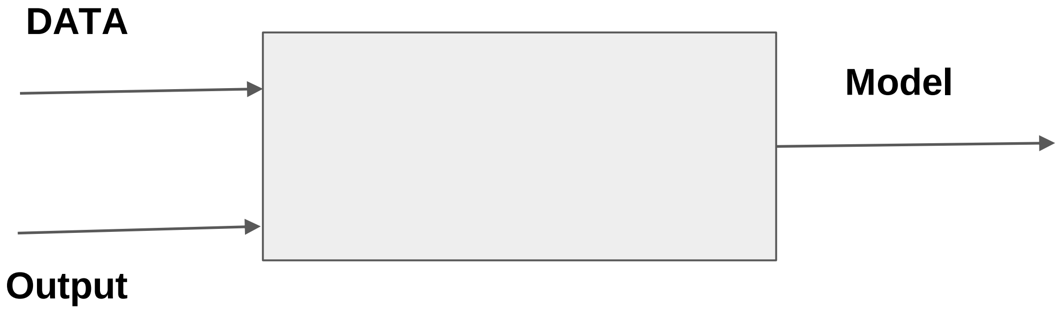
Traditionally



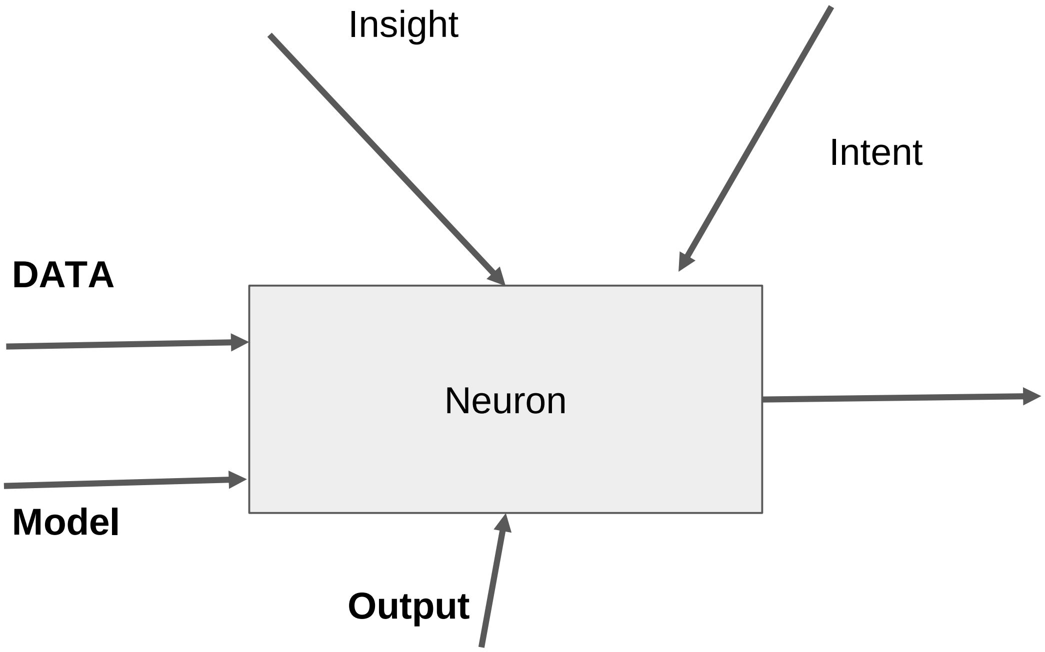
DATA

Output

Algorithm


DATA
Output
Model


DATA
Model
Predictions

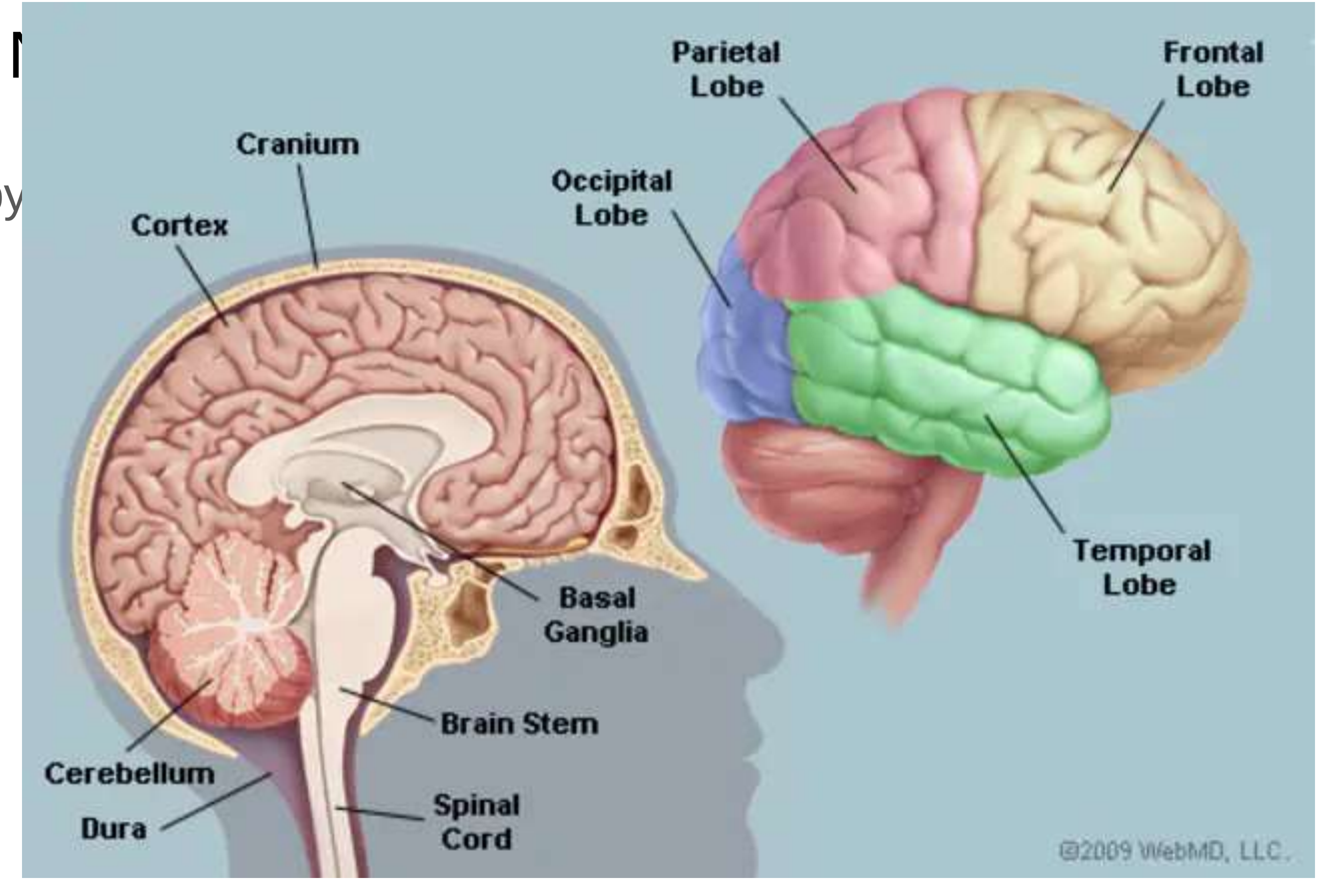

Insight
Intent
DATA
Neuron
Model
Output

Why do we need now?

- Data is Prevalent?
- Improved Hardware Architecture
- New Software Architecture

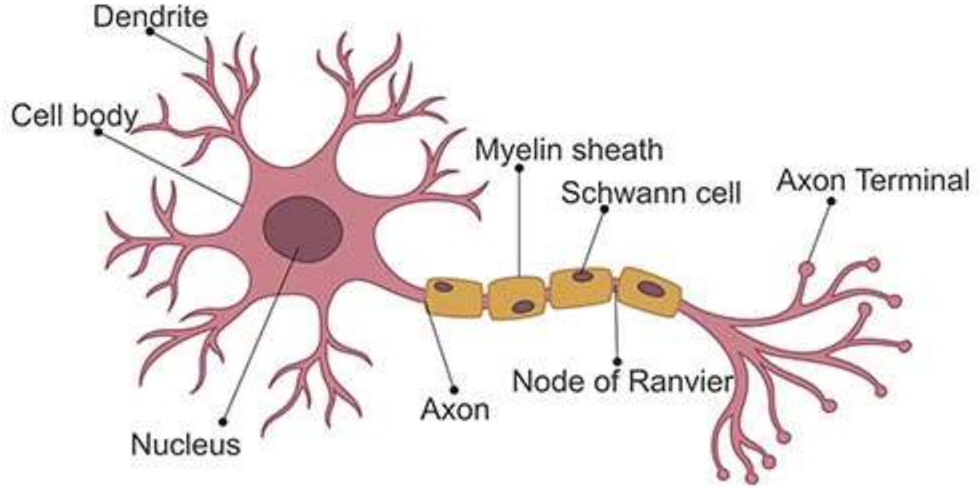
Neural N

Inspired by



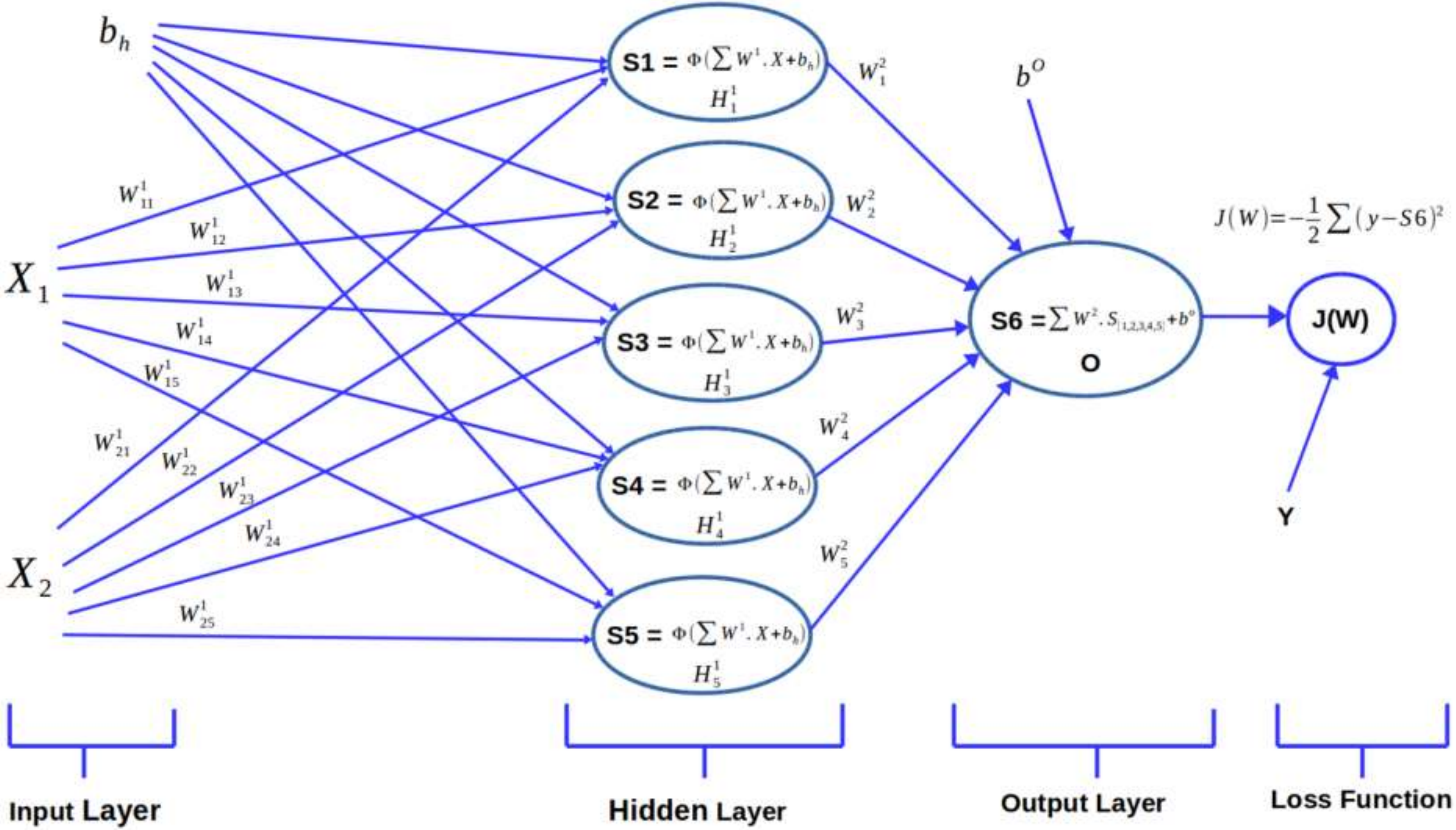
Building Block of the Neural Network is

Neuron



Neural Network

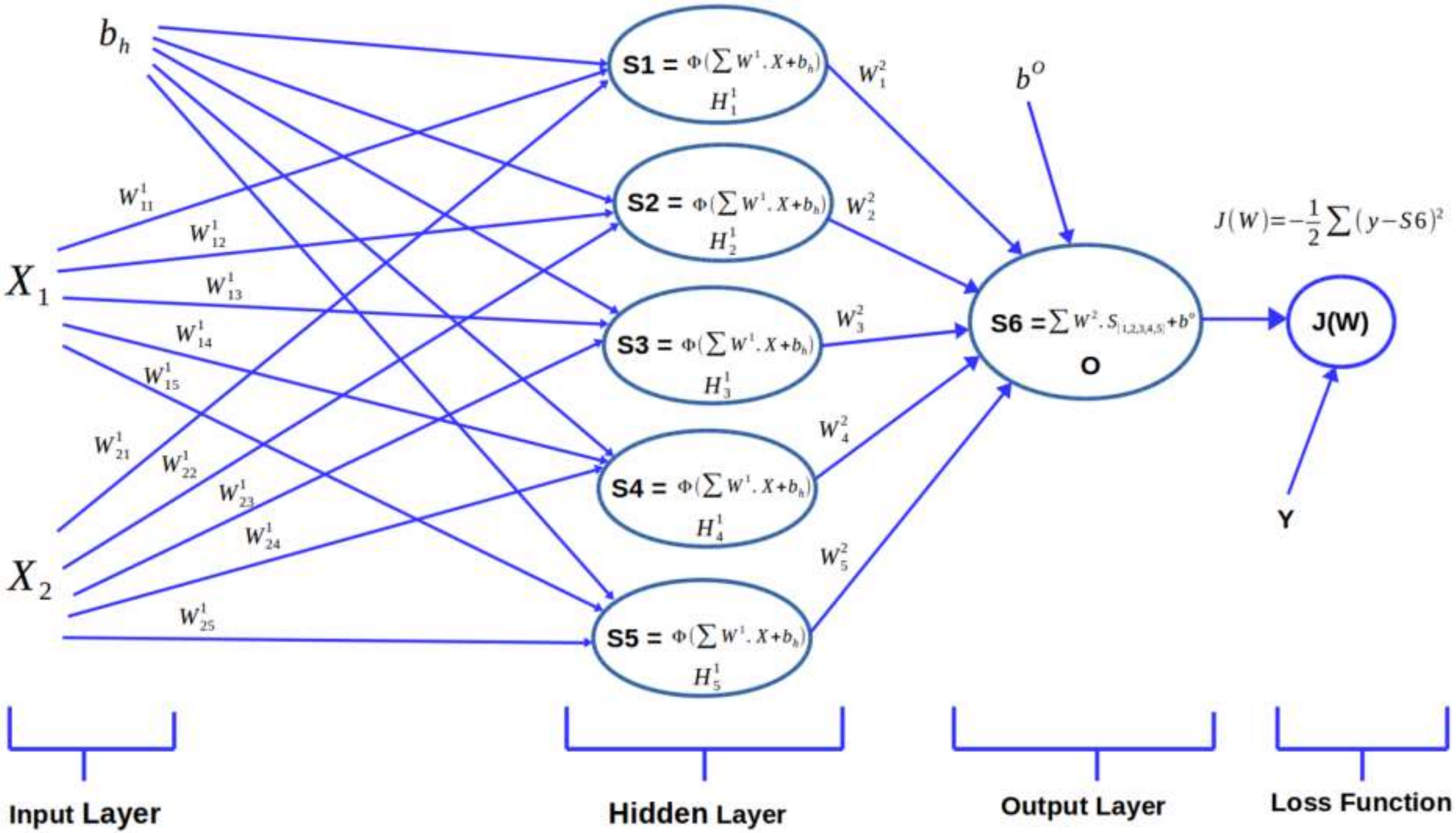
- Take data as the input
- Train themselves to understand the patterns in the data



b_h
X_1
X_2
W^1_{11}
W^1_{12}
W^1_{13}
W^1_{14}
W^1_{15}
W^1_{21}
W^1_{22}
W^1_{23}
W^1_{24}
W^1_{25}
S1 = $\Phi(\sum W^1 . X + b_h)$
H^1_1
S2 = $\Phi(\sum W^1 . X + b_h)$
H^1_2
S3 = $\Phi(\sum W^1 . X + b_h)$
H^1_3
S4 = $\Phi(\sum W^1 . X + b_h)$
H^1_4
S5 = $\Phi(\sum W^1 . X + b_h)$
H^1_5
W^2_1
W^2_2
W^2_3
W^2_4
W^2_5
b^O
S6 = $\sum W^2 . S_{[1,2,3,4,5]} + b^o$
O
$J(W) = -\frac{1}{2}\sum(y - S6)^2$
J(W)
Y
Input Layer
Hidden Layer
Output Layer
Loss Function

Learning Process of the Neural Network

1. Learning Process of the Neural Network
2. Forward Propagation
3. Back Propagation



b_h
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W^1_{11}
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H^1_3
S4 = $\Phi(\sum W^1 . X + b_h)$
H^1_4
S5 = $\Phi(\sum W^1 . X + b_h)$
H^1_5
W^2_1
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W^2_3
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J(W)
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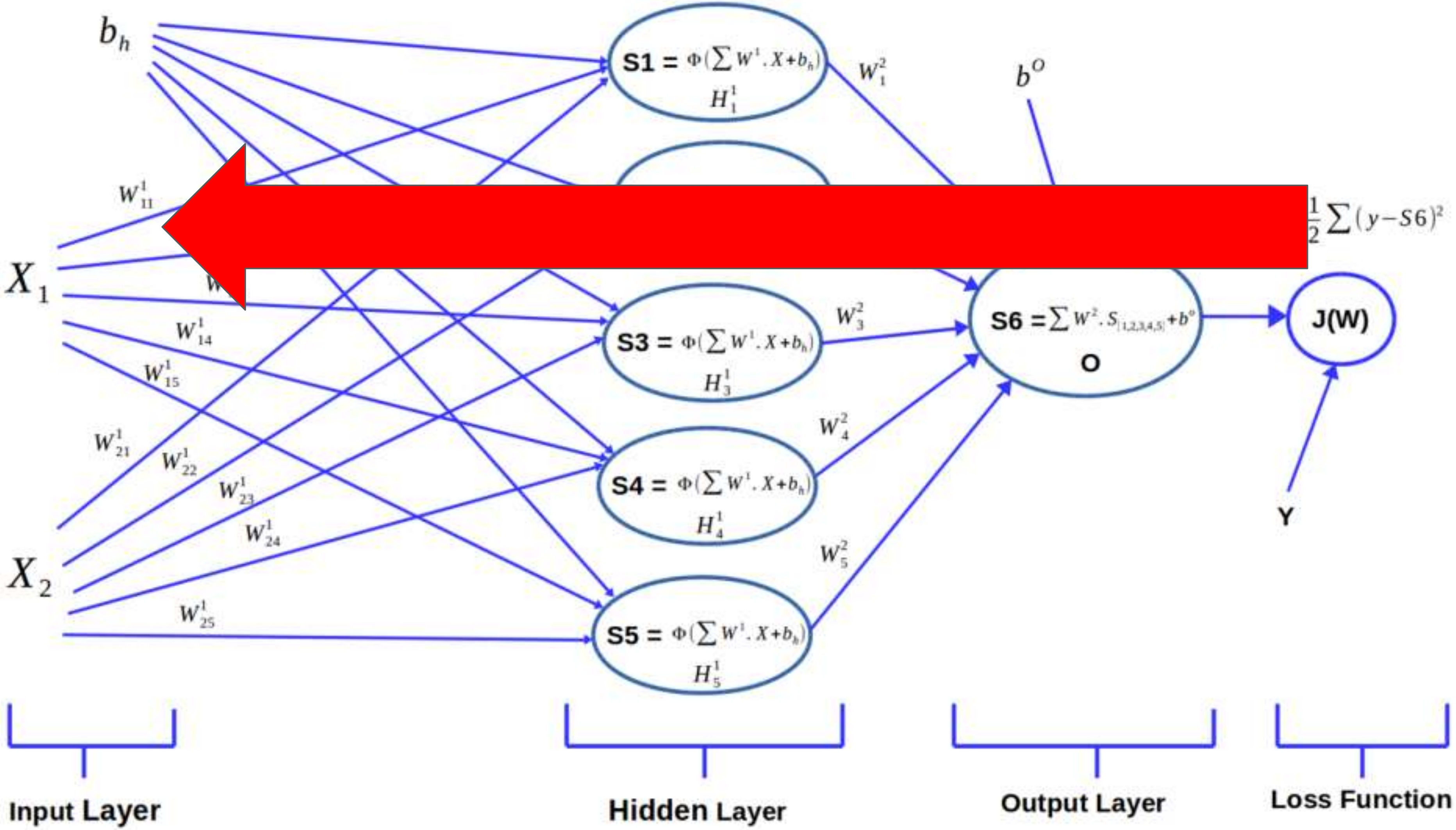
Input Layer
Hidden Layer
Output Layer
Loss Function

Weights and Biases

- **Weights -** How important information can this neural network can get
- **Bias -** allow the right decision to be taken into consideration



b_h
S1 = $\Phi(\sum W^1 . X + b_h)$
H_1^1
W_1^2
b^O
W_{11}^1
$\frac{1}{2}\sum(y-S6)^2$
X_1
W_{14}^1
W_{15}^1
S3 = $\Phi(\sum W^1 . X + b_h)$
H_3^1
W_3^2
S6 = $\sum W^2 . S_{[1,2,3,4,5]} + b^o$
O
J(W)
W_{21}^1
W_{22}^1
W_{23}^1
W_{24}^1
W_4^2
S4 = $\Phi(\sum W^1 . X + b_h)$
H_4^1
Y
X_2
W_5^2
W_{25}^1
S5 = $\Phi(\sum W^1 . X + b_h)$
H_5^1
Input Layer
Hidden Layer
Output Layer
Loss Function

In Backpropagation

Loss function helps the Neural Network Quantify the deviation front he expected output.

Randomly Initialize the Parameters

Back Propagation

- Use of the Loss Function
- Go Backwards and self tune the initial weights and biases
- Values adjusted to better fit prediction of the model that is trained from the data.

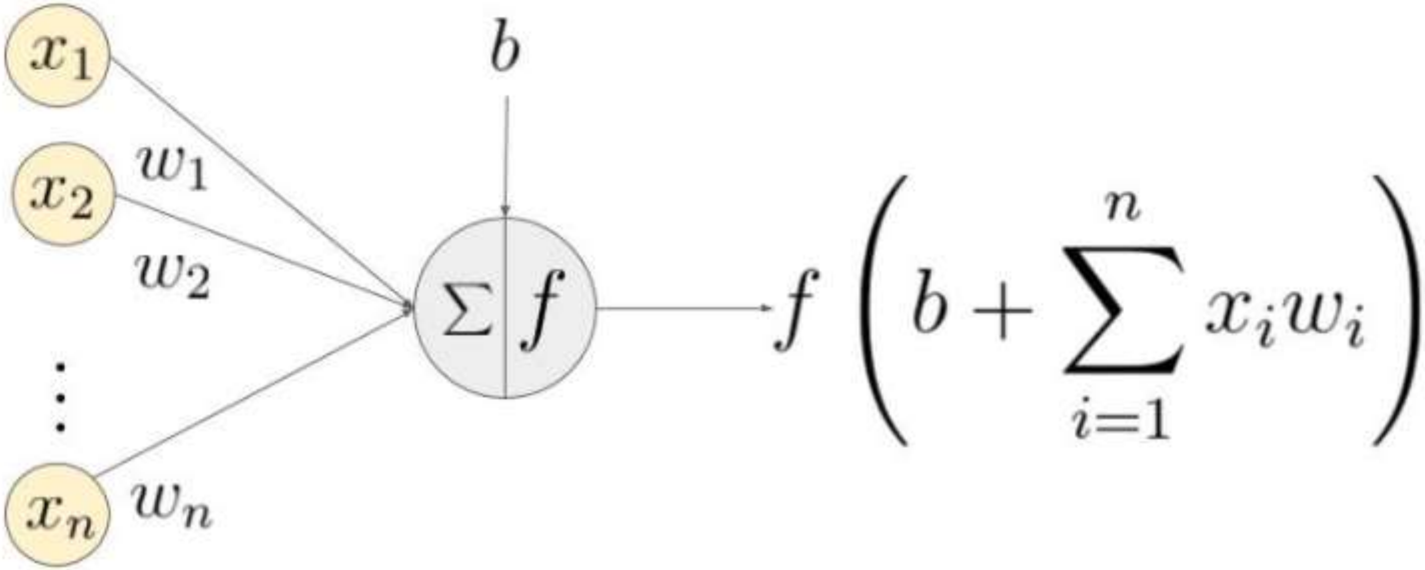
Learning Algorithm for the Neural Network

- Initialize the parameters with some tuned and calculated value.
- Feed Input data to the Network
- Compare the Predicted value with the expected data and calculate loss
- Perform Backpropagation to propagate this loss back through the network.
- Update parameters based on the loss
- Iterate the previous steps till the loss is minimized.

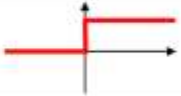
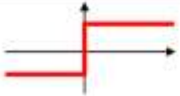
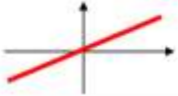
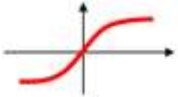
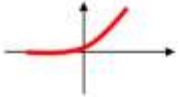
Terms used in the Neural Network

Activation Function

- Helps to decide whether a Neuron can be a drop out or can contribute to the next layer based on the dataset that it is trained on.
- Introduce non linearity into the Neural Network.



An example of a neuron showing the input (x_1 - x_n), their corresponding weights (w_1 - w_n), a bias (b) and the activation function f applied to the weighted sum of the inputs.

Activation function	Equation	Example	1D Graph
Unit step (Heaviside)	$\phi(z) = \begin{cases} 0, & z < 0, \\ 0.5, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Sign (Signum)	$\phi(z) = \begin{cases} -1, & z < 0, \\ 0, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Linear	$\phi(z) = z$	Adaline, linear regression	
Piece-wise linear	$\phi(z) = \begin{cases} 1, & z \geq \frac{1}{2}, \\ z + \frac{1}{2}, & -\frac{1}{2} < z < \frac{1}{2}, \\ 0, & z \leq -\frac{1}{2}, \end{cases}$	Support vector machine	
Logistic (sigmoid)	$\phi(z) = \frac{1}{1 + e^{-z}}$	Logistic regression, Multi-layer NN	
Hyperbolic tangent	$\phi(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	Multi-layer Neural Networks	
Rectifier, ReLU (Rectified Linear Unit)	$\phi(z) = max(0, z)$	Multi-layer Neural Networks	
Rectifier, softplus	$\phi(z) = \ln(1 + e^z)$	Multi-layer Neural Networks	

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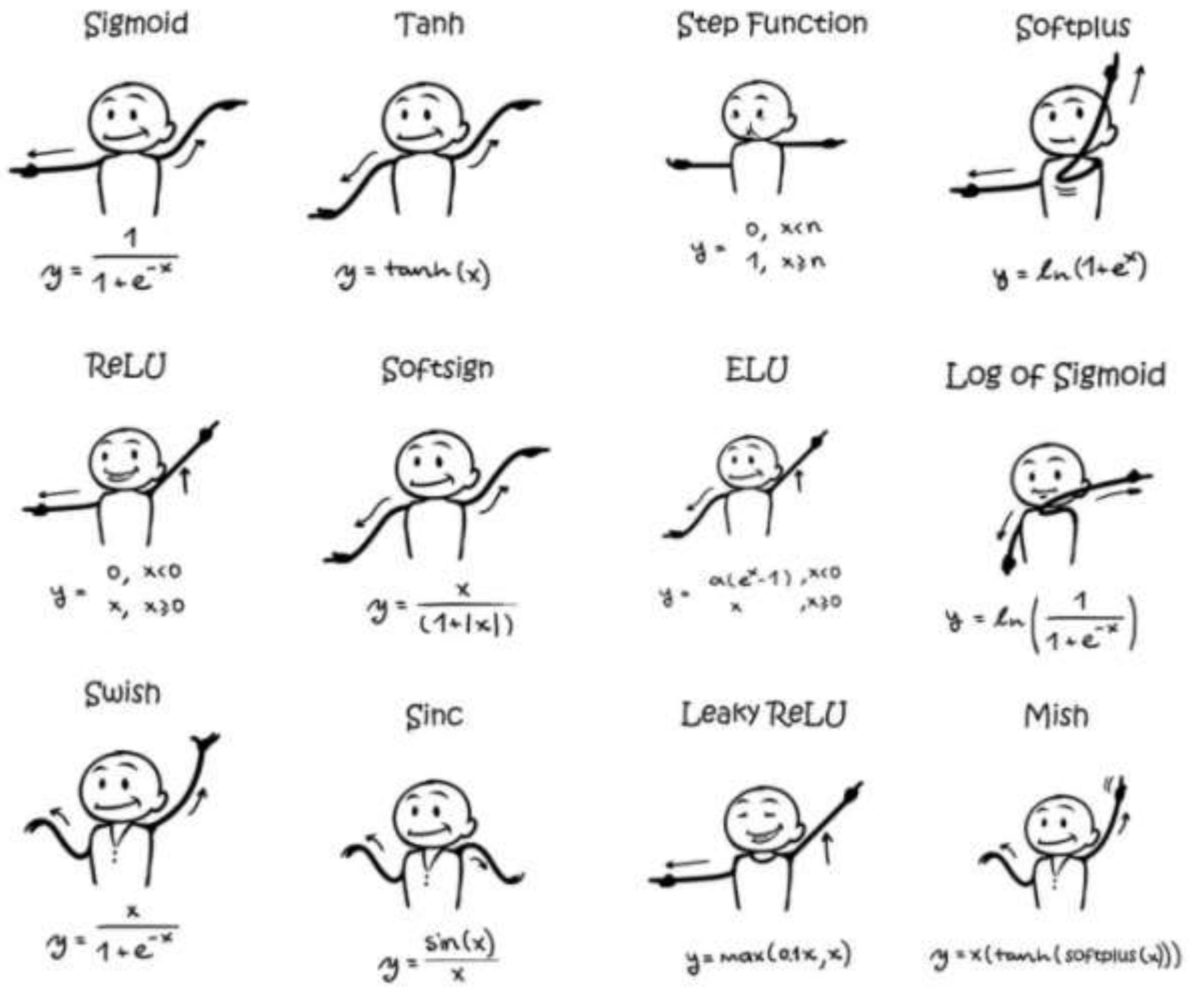


Sigmoid
$y = \frac{1}{1+e^{-x}}$
Tanh
$y = \tanh(x)$
Step Function
$y = 0,\ x<n$; $1,\ x\geq n$
Softplus
$y = \ln(1+e^{x})$
ReLU
$y = 0,\ x<0$; $x,\ x\geq 0$
Softsign
$y = \frac{x}{(1+|x|)}$
ELU
$y = \alpha(e^{x}-1),\ x<0$; $x,\ x\geq 0$
Log of Sigmoid
$y = \ln\left(\frac{1}{1+e^{-x}}\right)$
Swish
$y = \frac{x}{1+e^{-x}}$
Sinc
$y = \frac{\sin(x)}{x}$
Leaky ReLU
$y = \max(0.1x, x)$
Mish
$y = x(\tanh(\text{softplus}(x)))$

Which Activation Function to use?

- For the Binary Classification: Sigmoid or the Relu is Used for the best results.
- In the case of classifiers, sigmoid functions and their combinations often perform better.
- Because of the vanishing gradient issue, sigmoids and tanh functions are sometimes avoided.
- The ReLU function is a generic activation function that is employed in the majority of applications these days.

Activation Function Condition

- If we have dead neurons in our networks, the leaky ReLU function is the best option.
- Remember that the ReLU function should only be utilised in the hidden layers.
- As a general guideline, you should start with the ReLU function and then go on to other activation functions if the ReLU function does not produce the best results.

The king here is the data nad queen also

No matter what you are training a model using a dataset that is available for you,Neural Network is as it becomes.

Loss Function

We know that from the Random Weights and Biases the Neural Network Makes decision - Expected Output ,Weights and Biases are calculated.

Thus they Quantify the deviation of the predicted output by the NEural NEtwork to the expected output.

The loss functions in Regression are

- Absolute Error Loss
- Huber Loss
- Squared Error Loss

The loss Function in Binary Classification

Binary Cross Entropy

Hinge Loss

Multi Class Classification loss functions

Multi-Class Cross Entropy Loss

KL(Kullback–Leibler) Divergence

Optimizers

During the Training process we will adjust the parameters to minimize the loss function and make our model as optimized as possible for the Use.

Optimizers are basically

A function that combines to the loss function and model parameters by updating the Neural Network based on the output of the Loss Function.

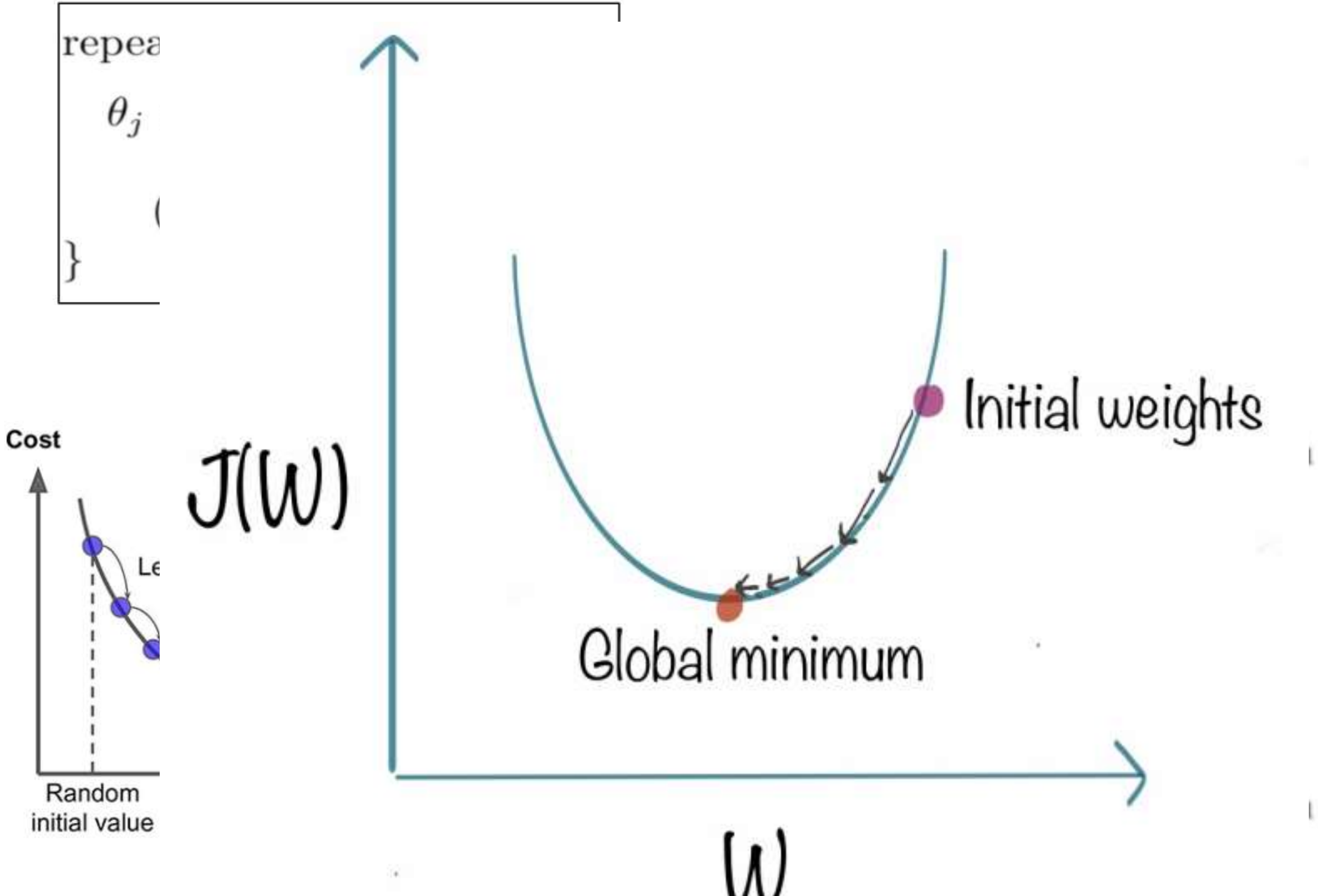
Gradient Descent

Iterative function that starts off at a random Point on the loss function and travel down its slope in steps(learning rate -from user) until it reaches the lowest point of the function.

1. Again this depends on the data.but they are
2. Most popular optimizer
3. Fast,Robust,Flexible.

Algorithm in the Lay man Terms

1. Calculate what a small change in the each individual weights would do to the loss function
2. Adjust each parameter based int eh its gradient(differential)
3. Repeat Steps one and Two until lower loss function is calculated by the Neural Network.



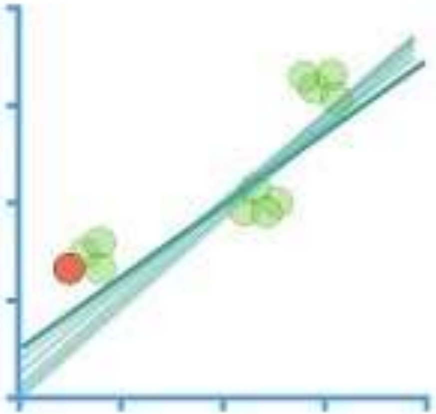
repea
θ_j
Cost
Random
initial value
J(W)
Initial weights
Global minimum
W

To avoid getting stuck in the local minima

We use Learning Rate

- Usually a small number that is multiplied to the scale of the gradients,which is any changes made to the weights are quite small.
- If we take large steps as learning rate then algorithm will tend to overshoot the global minimum
- Where we also don't want the algorithm to take forever to train and converge to the Global minimum.

Stochastic Gradient Descent…



…Clearly Explained!!!

They are more robust ,why?

- Like Gradient Descent,Except uses a subset of training example rather than the entire lot.
- SGD is Gradient descent that uses batch on each training.
- Use of the Momentum to Accumulate gradients.
- Less intensive computation as they are batched.

Backpropagation

A simple implementation of the Gradient descent on the neural network.

AdaGrad

- Adaptive learning rate to individual features.
- Some weights will have different learning rates
- Ideal for the Sparse datasets with many input examples missing
- Learning rate tends to get lower accordingly.

Parameters and Hyperparameters

- What are model parameters?
- Variable from the neural Network whose values can be estimated from the data.
- Required by the model to make prediction
- Value define the learnt parameters from the data.
- Not set manually.saved as the Neural Network is trained.

Example - Weights and Biases.

What are model Hyper parameters?

- They are configured externally to the neural network : Value cannot be estimated until we train the dataset on the neural network
- No clear way to find the best value
- When the DL algorithm is tuned ,you are really tuning the hyperparameters.
- This is tuned manually

Example - Learning Rate,C and Alpha in SVM,Epochs,k in the kNEes

Summary

Model parameters -> Estimated front the data

Model Hyperparameters -> Can't be estimated from the data

HyperParameters are often called as the parameters as they are a part of the Machine Learning that must set manually and tuned.

Epochs,Batches,Batch Size and Iterations

Need to learn to do this to your Neural Network when the dataset is too Big.

Break the dataset into smaller chunks and feed those chunks to the Neural Network One by One.

Epochs

When the Entire dataset is passed forward to the Neural Network and only once they get trained in the Network.

We use more than one epoch to help model generalize better and accurate.

There is no absolute count for the dataset as its different for different datasets.

Batch and Batch size

We divide large dataset into the smaller batches and feed those batches to the Neural Network

Batch Size - Total number of the training examples in the Batches.

Iterations

Number of Batches needed to complete one epoch

Number f batches = Number of iterations in one epochs

Let's have some more insights

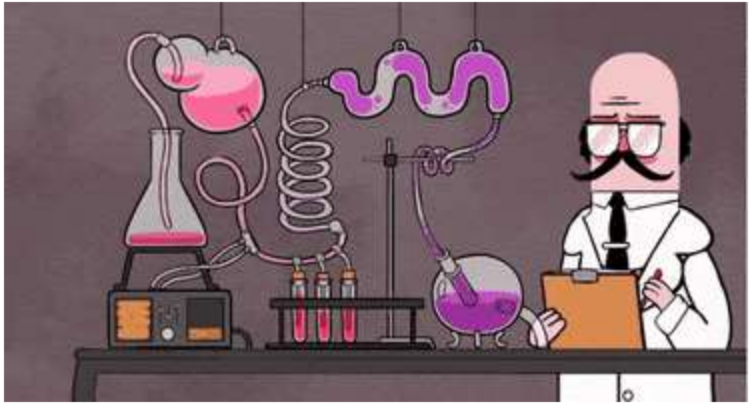
Suppose we have 1 million Number if dataset as the Training Example and you divide the dataset into the batches of 500 ,to Complete 1 Epoch ,it would take 20000 iterations.

Conclusion for the terms used in NN’s

How to design an Architecture?

Which Activation Function to use?

The only thing is to



Types of Learning

There are Three Main Types -

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Supervised Learning

- Algorithms designed to learn from Examples.
- Models are trained on well-labelled data
- Each example has
- Input Object - Typically a Vector
- Desired Output Value Supervised Signal

During Training

Searches the pattern and correlate with the desired output.

After Training

Takes the unseen inputs and determine which label to classify it to.

Objective of a Supervised learning model

Is to predict the correct label for the unseen data.

$$y=f(x)$$

Supervised learning is of two types

- Classification
- Regression

Classification

- Take Input data and assign it to a class/category.
- Models finds features in the data that correlates to the class and creates a mapping function
- This mapping function will be used to classify unseen data from testing and the validation set from the cross validation of the data

Binary and Multiclass classification

Definition and Example

Popular Classification Algorithms

- Logistic Regression.
- Naïve Bayes.
- Stochastic Gradient Descent.
- K-Nearest Neighbours.
- Decision Tree.
- Random Forest.
- Support Vector Machine

Regression

Model tries to find a relationship between dependent and independent variable.

Goal is always to predict continuous values such as a test score.

Equation is always continuous

predictor, 'x-variable', independent variable, explanatory variable

coefficient

$$Y = \underbrace{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p}_{\text{linear predictor}} + \varepsilon$$

response, dependent variable, observation, 'y-variable'

random error, "noise"

Simple Linear Regression

$$y = W x + b$$

Different Regression Algorithm

- Linear **Regression**.
- Logistic **Regression**.
- Ridge **Regression**.
- Lasso **Regression**.
- Polynomial **Regression**.
- Bayesian Linear **Regression**.

Application of Supervised Learning

- Text categorization
- Face Detection
- Signature recognition
- Customer discovery
- Spam detection
- Weather forecasting
- Predicting housing prices based on the prevailing market price
- Stock price predictions, among others

Unsupervised Learning

- Uses to manifest underlying pattern in data
- Used in Exploratory Data Analysis
- Need no labelled data,they use the feature from the Data

Unsupervised Learning is of

- Clustering
- Association

Clustering -Partitional Clustering

- Partitional Clustering
- Each Data point can belong to a single cluster

Clustering - Hierarchical Clustering

- Clusters within the clusters
- Datapoint may belong to different clusters

Association

Attempts to find different relationship between the different entities.

Example - Market Basket Analysis

Some Clustering Algorithm

1. Clustering Dataset
2. Affinity Propagation
3. Agglomerative Clustering
4. BIRCH
5. DBSCAN
6. K-Means
7. Mini-Batch K-Means
8. Mean Shift
9. OPTICS
10. Spectral Clustering
11. Gaussian Mixture Model

Application of the Unsupervised Learning

- Fraud detection
- Malware detection
- Identification of human errors during data entry
- Conducting accurate basket analysis, etc.

Reinforcement Learning

Enable the intelligent entity to learn in an interactive environment by trial and error(by Policy and reward network) based on its own actions and experience.

This is a very new way of getting the things learnt.

If your Neural Network doesn't work well then you have to use the Reinforcement :Learning.

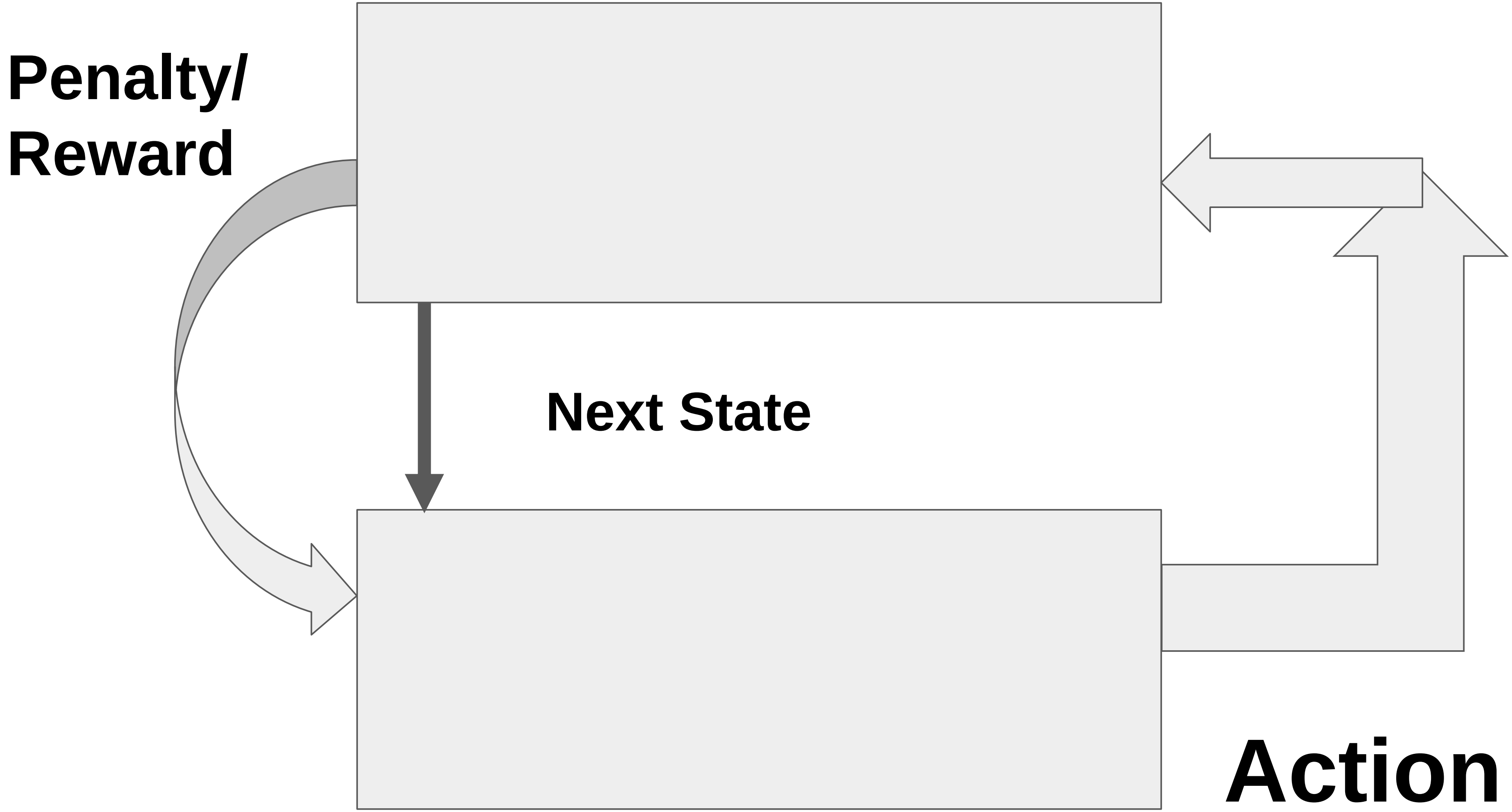
Reward a

Uses the pos what has been learnt.

Goal of Reinforcement learning is to

- Find a Suitable model that would maximize the total cumulative reward and make a very approximate result that might help in making some more discussion.
- Maximize the points won in a training over many examples
- Penalize when they make wrong decisions
- Reward where they make Right decision

Usually modelled as a “Markov Decision Process”


Penalty/
Reward
Next State
Action

Application of the Robotics

- Robotics
- Business strategy
- Traffic Light Control
- Web system configuration
- NLP
  - to personalize suggestions
  - deliver more meaningful notifications to users
  - optimize video streaming quality.
- Gaming
- Bidding

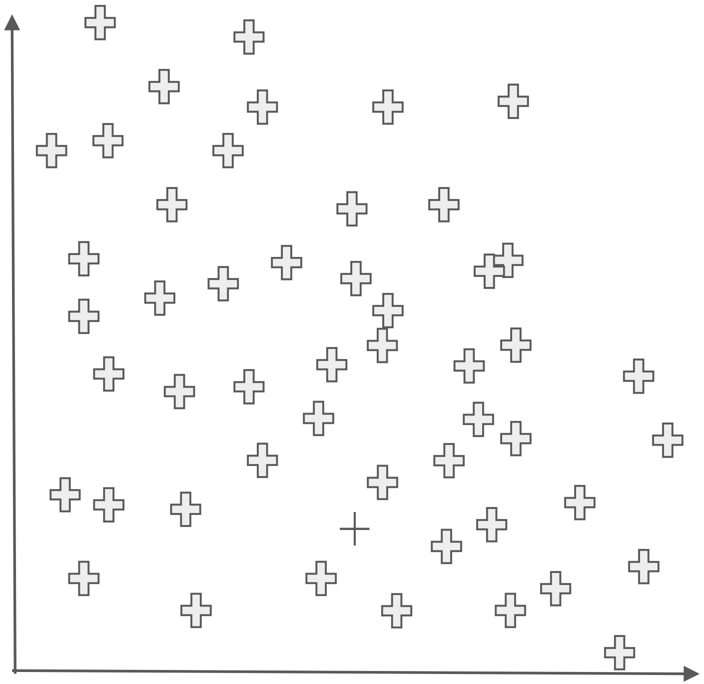
Some core and Canonical Problems in Deep Learning

Basically we find this as the situation:

Model should perform well on training data and new test data

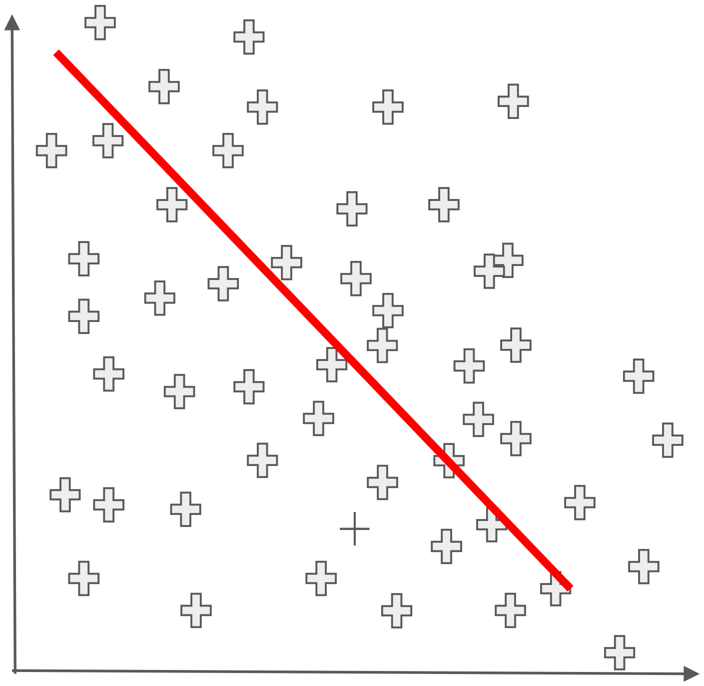
Most common problem faced will always be overfitting

So the data points as the example is here

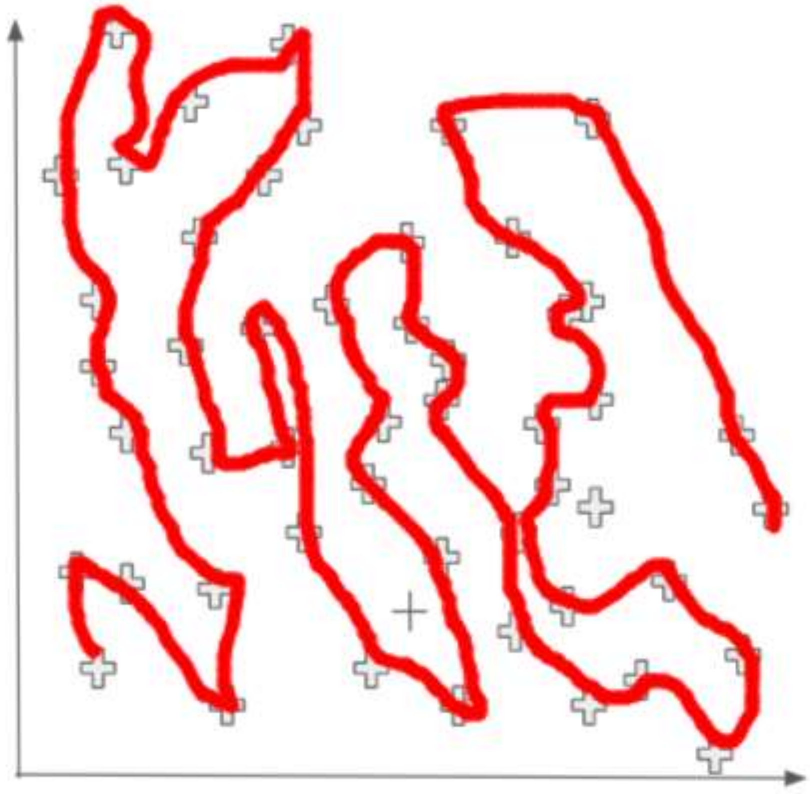


- Data is Skew
- Data is Random
- They don't care anything and anybody they are just generated
- They are collected to make sense and make a model
- They are collected to make the right decision from the model

Underfitting

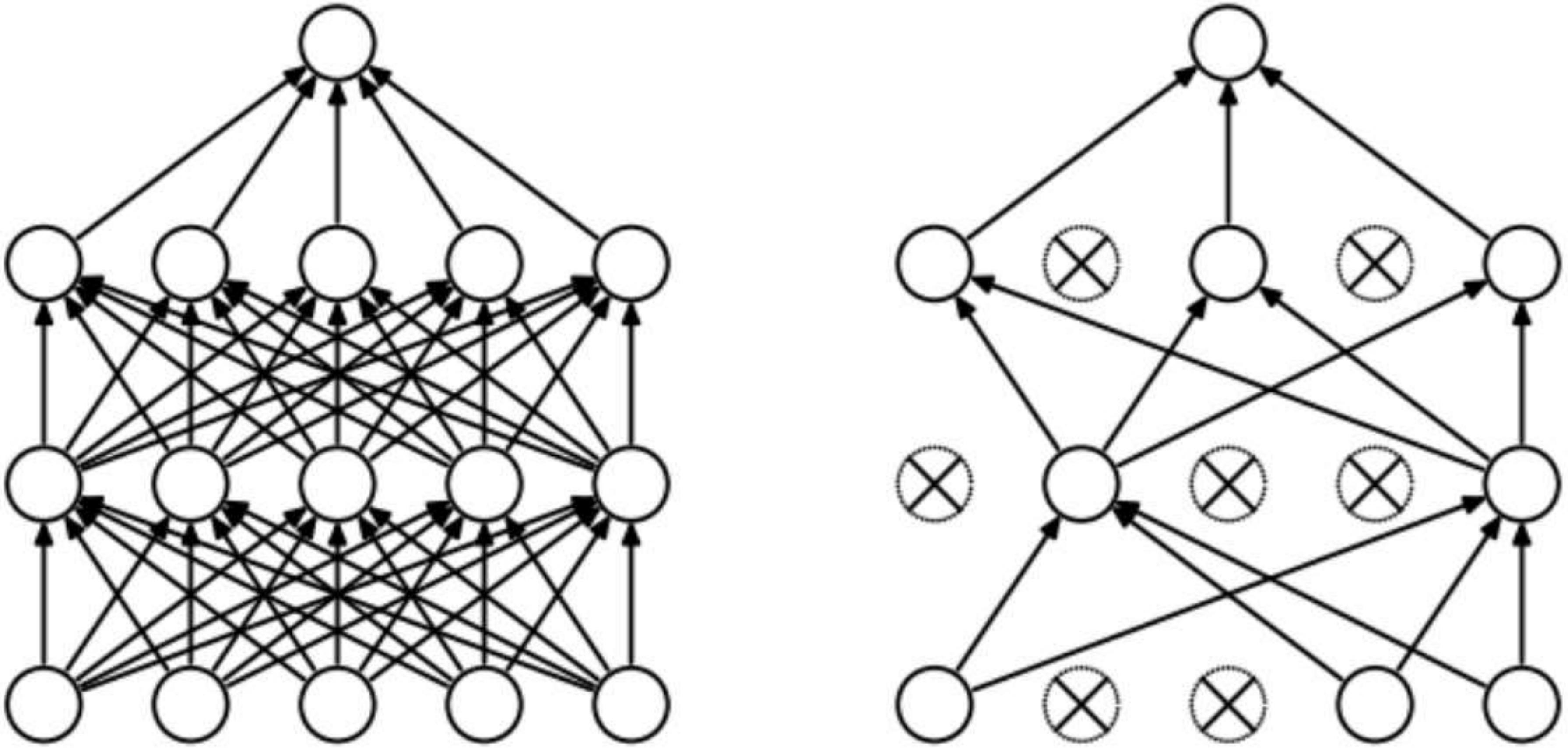


Over-fitting



Tackling Overfitting is

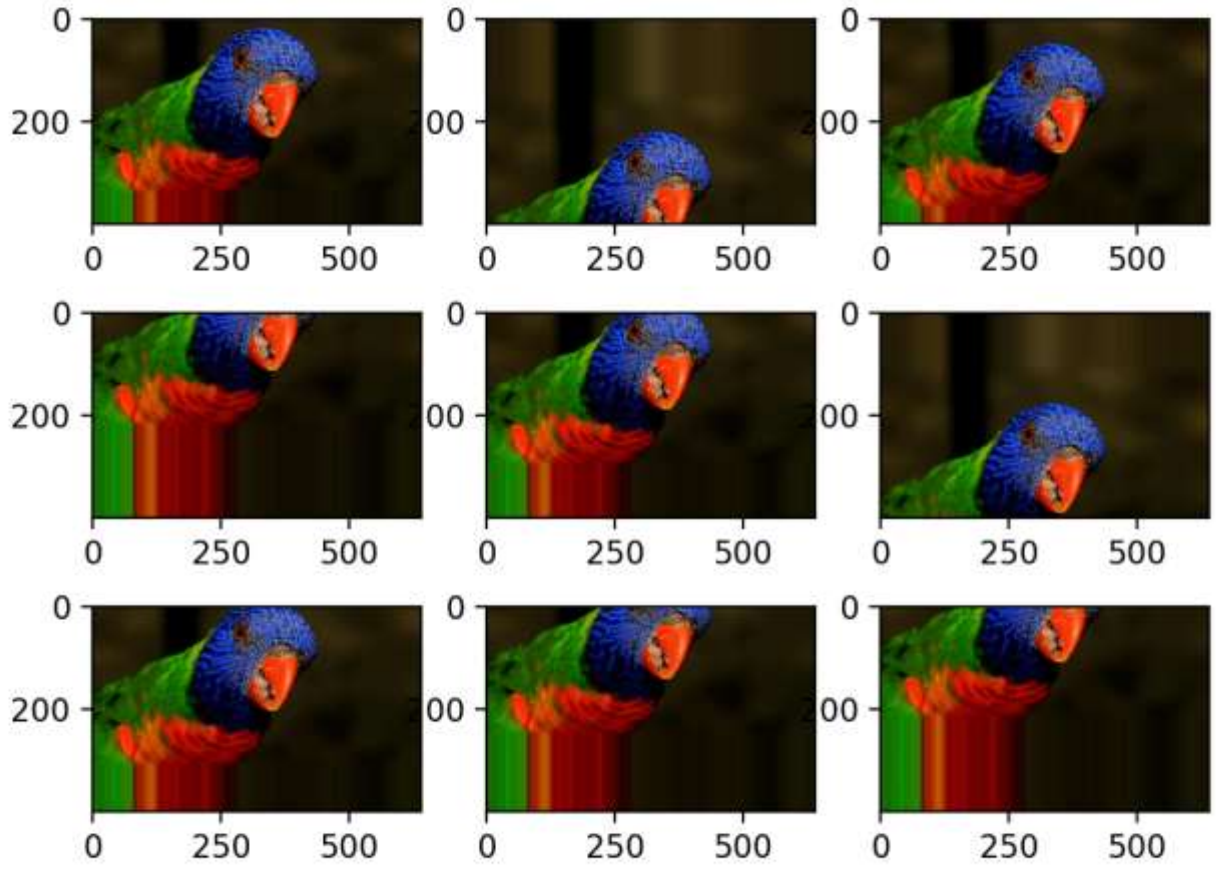
1. Hold-out
2. Cross-validation
3. Data augmentation
4. Feature selection
5. L1 / L2 regularization
6. Remove layers / number of units per layer
7. Dropout
8. Early stopping



(a) Standard Neural Net (b) After applying dropout.

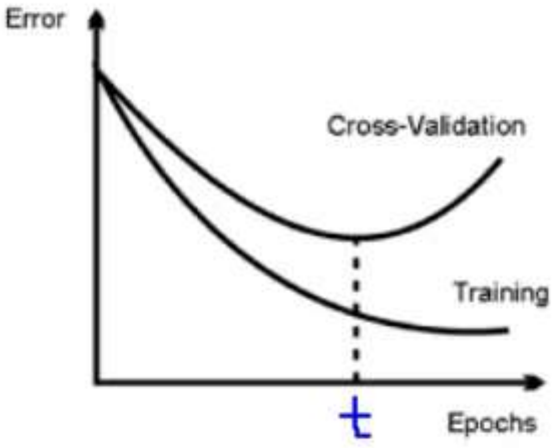
Data Augmentation

Just create some fake data as much as possible from the data itself.


0
200
0
250
500

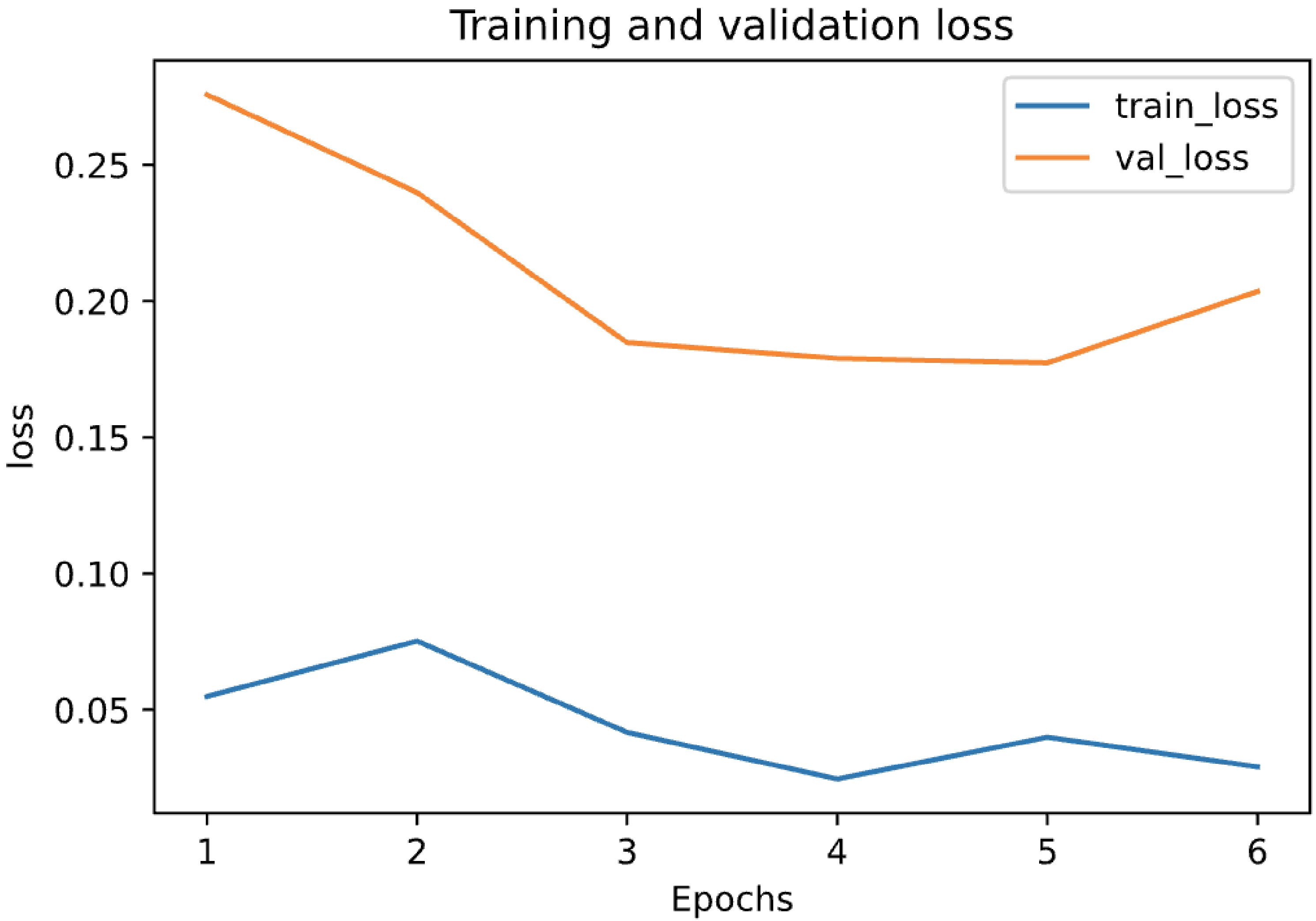
Early Stopping

[Use Early Stopping to Halt the Training of Neural Networks At the Right Time (machinelearningmastery.com)]



Whe

Train tain
point.



Neural Network are in plenty

So go the sources that I am saying in this stream.

So we talked a lot about the models

So now let's get to know how we can build a model.

Gathering Data

Picking the right data is very important,Good way to start is you need to make assumption about the data that you need.

Size of the data set also matters

No one size fits all

Amount of the data needed = 10 times the model parameters

Quality of the data also matters

Data has to be more Accurate and Reliable with no Adversaries.

Noiseless Features.

Some Dataset Repositories are

I will list out here.

Pre- Processing dataset

Split the dataset into the subset.

Training data set

Testing Data Set

Validation Data Set

We can randomly split the dataset

This process depends on

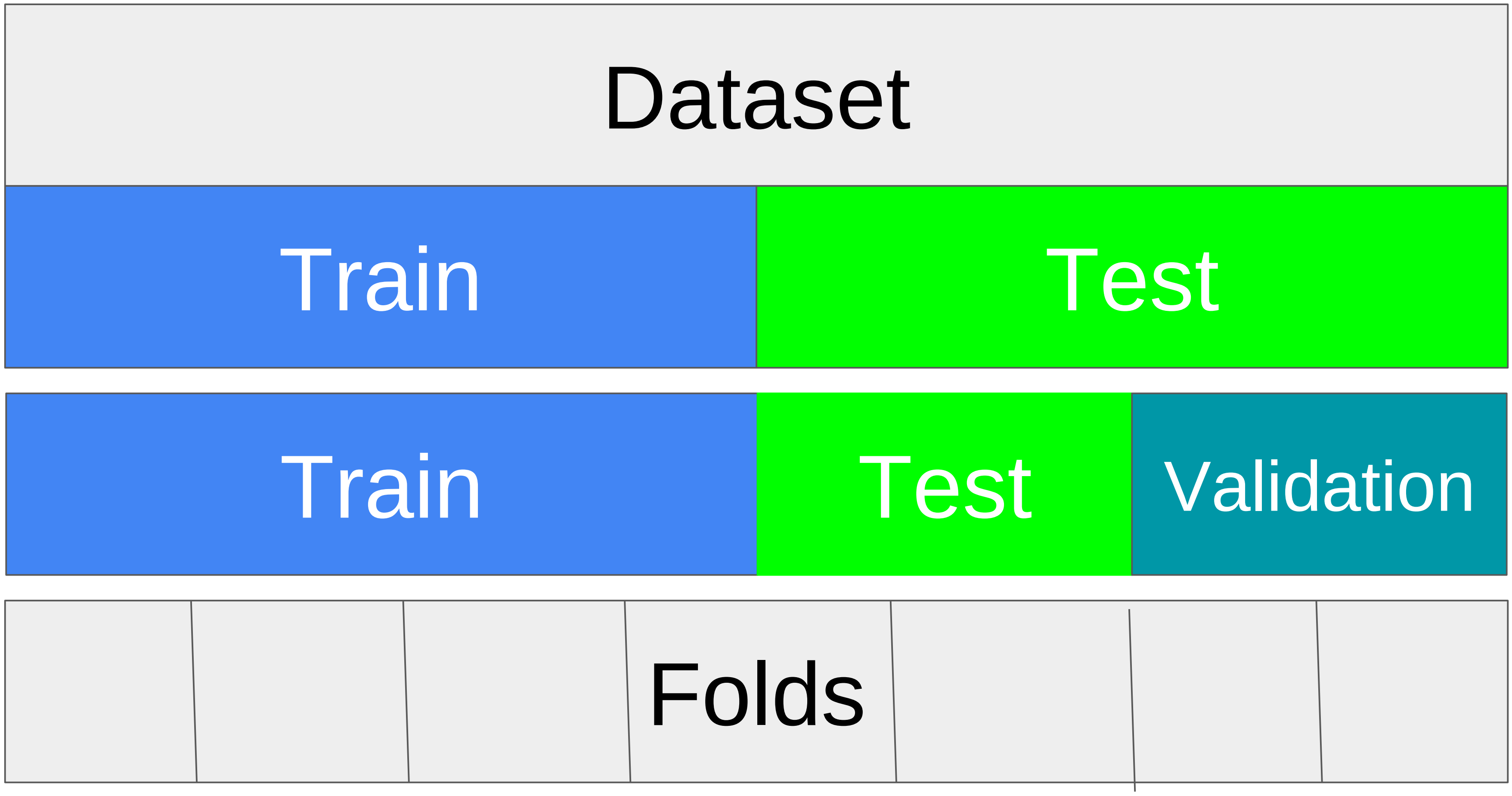
- Number of the samples in eh data
- Model Being Trained

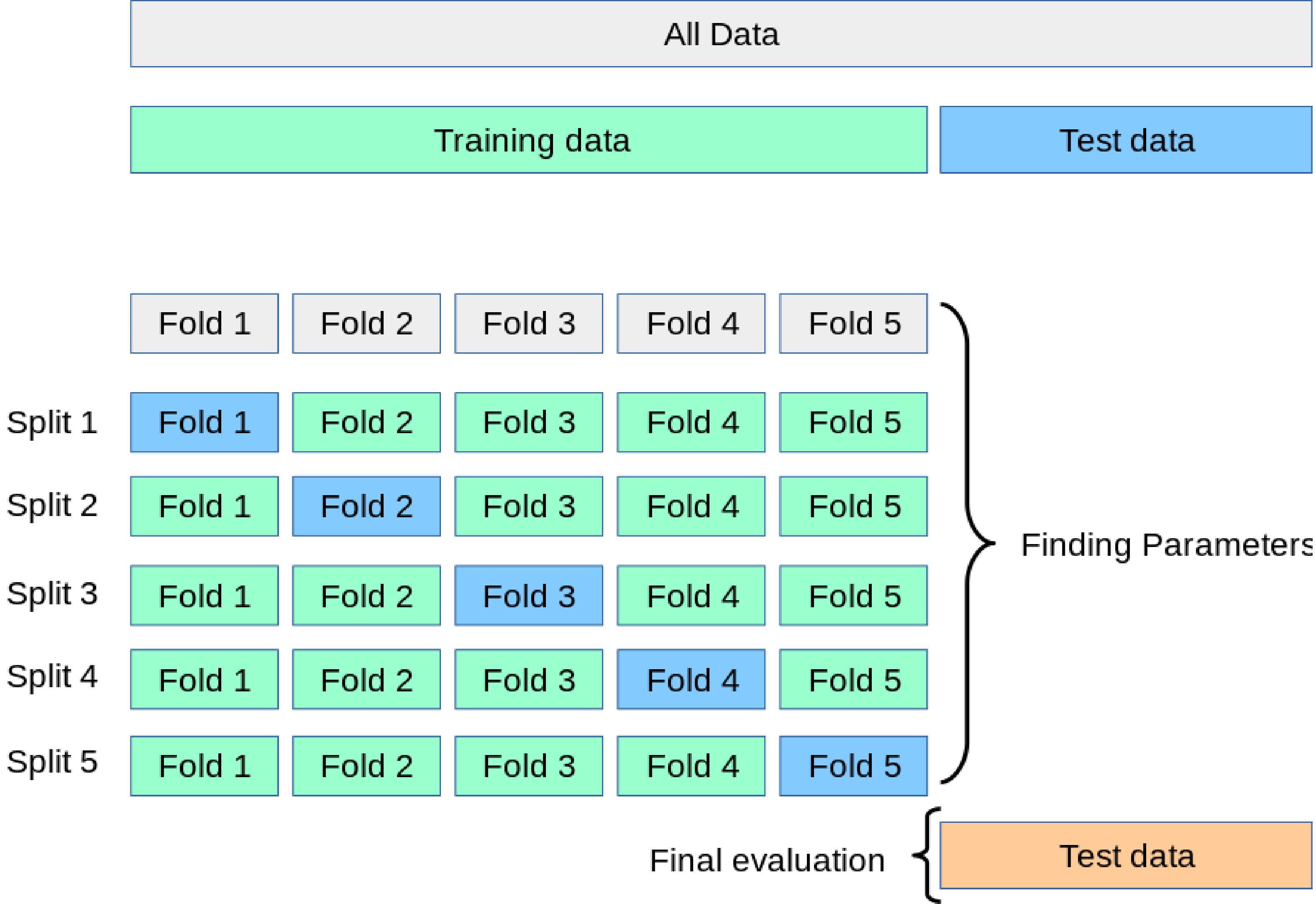
Simple rule of thumb

- Few Hyperparameters means small validation set
- Many hyperparameters means large validation set

The ratio in which you split the dataset is specific to your

Use Case


Dataset
Train
Test
Train
Test
Validation
Folds


All Data
Training data
Test data
Fold 1 Fold 2 Fold 3 Fold 4 Fold 5
Split 1 Fold 1 Fold 2 Fold 3 Fold 4 Fold 5
Split 2 Fold 1 Fold 2 Fold 3 Fold 4 Fold 5
Split 3 Fold 1 Fold 2 Fold 3 Fold 4 Fold 5
Split 4 Fold 1 Fold 2 Fold 3 Fold 4 Fold 5
Split 5 Fold 1 Fold 2 Fold 3 Fold 4 Fold 5
Finding Parameters
Final evaluation
Test data

Look for the missing data

- Nan or Null
- Eliminated Features or the Missing Value
- Impute the Missing data

Sampling

Use a sample of the dataset

Why we need this

- Faster Convergence
- Reduces the Disk Space

Preprocessing is Required for the FEature Scaling

- Crucial Step for the Model TRaining:
- Normalization
- Standardization

Then obviously train and Evaluate.

Optimization

Will be Continued ...